



Credit risk III.

Risk Management in Financial Institutions

Lecture 12

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Credit VaR

- Can be defined analogously to Market Risk VaR
- A one year credit VaR with a 99.9% confidence is the loss level that we are 99.9% confident will not be exceeded over one year
- Creditmetrics/KMV
- CreditRisk+

Credit VaR methodologies

- Structural/„Mark-to-Market” models
 - There is an economic model behind default (Merton)
 - Simpler normality assumptions: Vasicek ((IRB) model)
 - Full rating migration matrix estimate: CreditMetrics
- „Default mode”/”Actuary” models
 - The cause of default is not observable, it is like a damage event
 - Recursive summation of elementary distributions
 - Requires PD standard deviation in addition to PDs: CreditRisk+

Vasicek's Model (Equation 21.1, page 448)

- For a large portfolio of loans, each of which has a probability of PD of defaulting by time T the worst case default rate that will not be exceeded at the $X\%$ confidence level is

$$\text{WCDR} = N \left\{ \frac{N^{-1}[\text{PD}] + \sqrt{\rho} N^{-1}(X)}{\sqrt{1 - \rho}} \right\}$$

where ρ is the Gaussian copula correlation

$$\text{VaR} = \text{WCDR} \times \text{EAD} \times \text{LGD}$$

Vasicek or IRB model – more in detail (1)

(Radnai-Vonnák pp. 125-127., Hull 1. ed. pp. 159-160.)

$$\frac{V_1 - V_0}{V_0} = R : N(\mu, \sigma)$$

Assume that the value of a company's debt at the start is D_0 and the value of its shares is E_0 . Then obviously $V_0 = D_0 + E_0$. Assume that the interest on the debt is r using a simple yield calculation.

The company goes bankrupt if $V_1 \leq D_1$, i.e., $R \leq \frac{D_0(1+r)}{V_0} - 1 \quad Z_i$

The return on assets of company i . will be determined by the following relationship:

$$R_i = \sqrt{w_i} X + \sqrt{1 - w_i} \varepsilon_i$$

It can be seen that the expected value of R_i is zero, its standard deviation is 1, and since R_i is the sum of two independent normally distributed variables, R_i itself has a standard normal distribution!

where X is the yield of a so-called systematic risk factor (e.g. a stock market index that represents the movements of the whole market) with a standard normal distribution, and ε_i is the yield of a company-specific risk factor, also with a standard normal distribution, and X and ε are independent.

$$r(R_i, X) = \frac{\text{cov}(R_i, X)}{\text{var}(R_i) \text{var}(X)} = \frac{\text{cov}(\sqrt{w_i} X + \sqrt{1 - w_i} \varepsilon_i, X)}{1} = w_i \text{cov}(X, X) + (1 - w_i) \text{cov}(\varepsilon_i, X) = w_i$$

Merton
(1974)

Vasicek
(1987)

Vasicek or IRB model – more in detail (2)

$$PD_i = P(R_i < Z_i) = \Phi(Z_i) \Rightarrow Z_i = \Phi^{-1}(PD_i)$$

Worst Case Default Rate

$$P_i(X_{99,9\%}) = P(R_i(X_{99,9\%}) < Z_i) \quad R_i(X_{99,9\%}) = \sqrt{w_i} X_{99,9\%} + \sqrt{1-w_i} \varepsilon_i < Z_i \quad P_i(X_{99,9\%}) = \Phi\left(\frac{Z_i - \sqrt{w_i} X_{99,9\%}}{\sqrt{1-w_i}}\right)$$

$$P_i(X_{99,9\%}) = \Phi\left(\frac{\Phi^{-1}(PD_i) - \sqrt{w_i} \Phi(0,999)}{\sqrt{1-w_i}}\right)$$

If the individual size of each loan is negligible in terms of the size of the portfolio, the loss amount is equal to the hypothetical expected value of the loss.

Gordy
(2004)

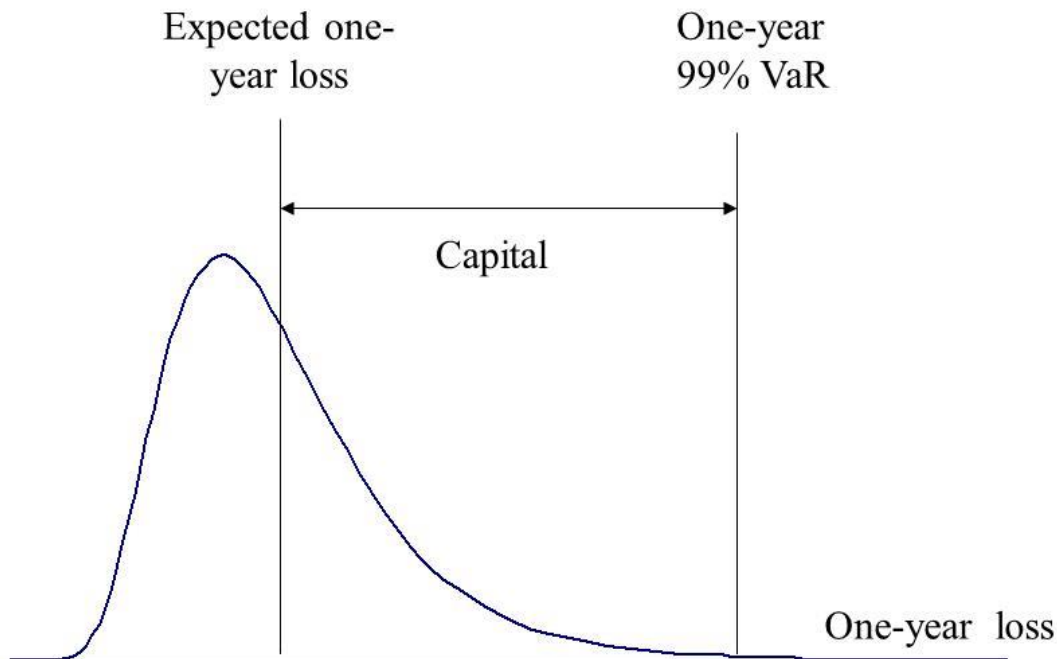
LGD is constant, independent of the return on the company's assets

$$L_i = EAD_i LGD_i P_i(X_{99,9\%})$$

Loan portfolio
VaR
= Basel 2
regulatory
capital
function

$$L = EAD_i LGD_i \Phi\left(\frac{\Phi^{-1}(PD_i) - \sqrt{w_i} \Phi^{-1}(0,999)}{\sqrt{1-w_i}}\right)$$

The Model used by Regulators (Figure 15.1, page 340)



Key Model (Gaussian Copula)

The 99.9% worst case default rate is

$$WCDR = N \left[\frac{N^{-1}(PD) + \sqrt{\rho} \times N^{-1}(0.999)}{\sqrt{1-\rho}} \right]$$

Numerical Results for WCDR

Table 15.5, page 342

	$PD=0.1\%$	$PD=0.5\%$	$PD=1\%$	$PD=1.5\%$	$PD=2\%$
$\rho=0.0$	0.1%	0.5%	1.0%	1.5%	2.0%
$\rho=0.2$	2.8%	9.1%	14.6%	18.9%	22.6%
$\rho=0.4$	7.1%	21.1%	31.6%	39.0%	44.9%
$\rho=0.6$	13.5%	38.7%	54.2%	63.8%	70.5%
$\rho=0.8$	23.3%	66.3%	83.6%	90.8%	94.4%

Dependence of ρ on PD

- For corporate, sovereign and bank exposure

$$\rho = 0.12 \times \frac{1 - e^{-50 \times PD}}{1 - e^{-50}} + 0.24 \times \left[1 - \frac{1 - e^{-50 \times PD}}{1 - e^{-50}} \right] = 0.12[1 + e^{-50 \times PD}]$$

PD	0.1%	0.5%	1.0%	1.5%	2.0%
$WCDR$	3.4%	9.8%	14.0%	16.9%	19.0%

(For small firms ρ is reduced)

Capital Requirements

$$\text{Capital} = \text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA}$$

$$\text{where MA} = \frac{1 + (M - 2.5) \times b}{1 - 1.5 \times b}$$

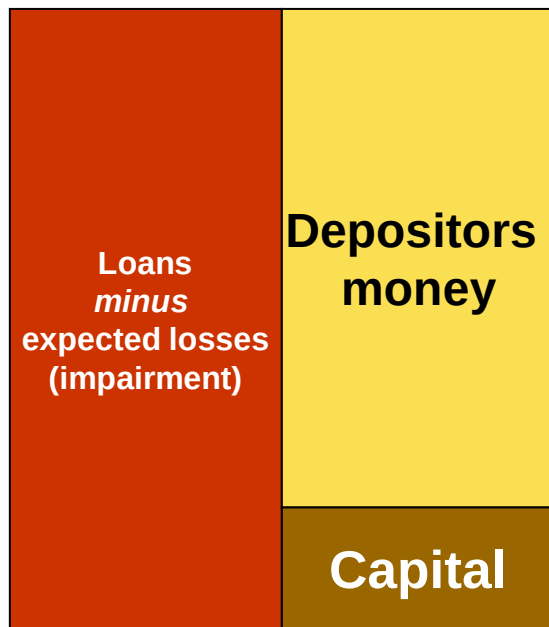
M is the effective maturity and

$$b = [0.11852 - 0.05478 \times \ln(PD)]^2$$

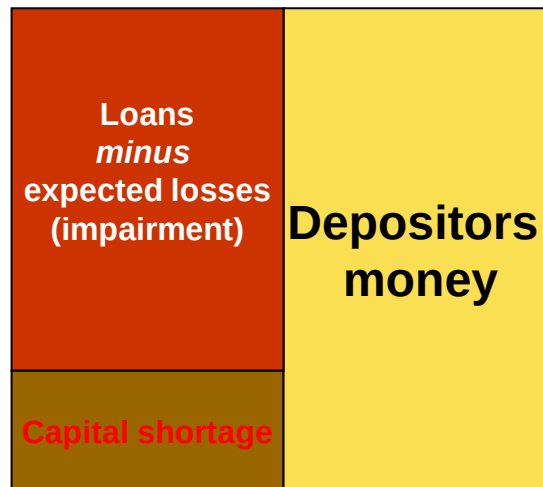
The risk - weighted assets are 12.5 times the Capital
so that Capital = 8% of RWA

Credit VaR and capital

OPERATING BANK



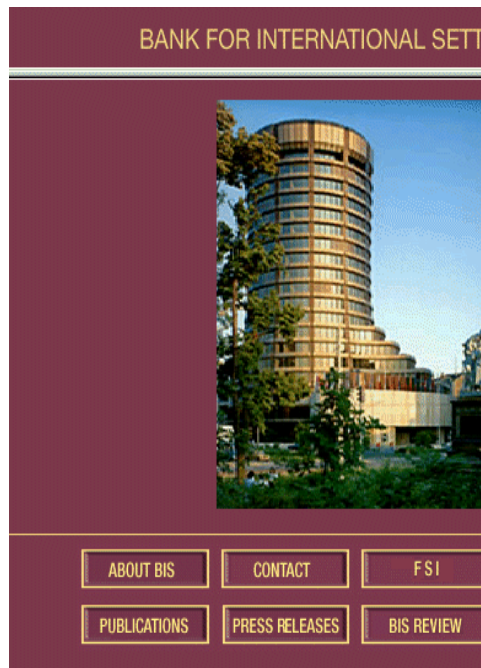
BANKRUPT BANK



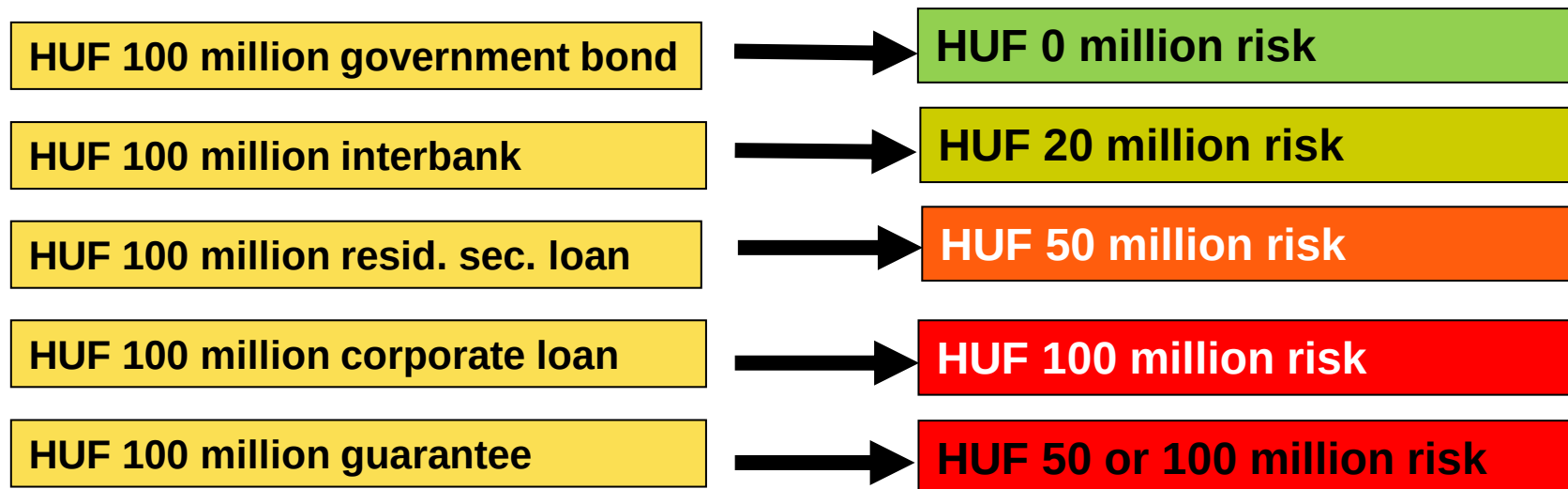
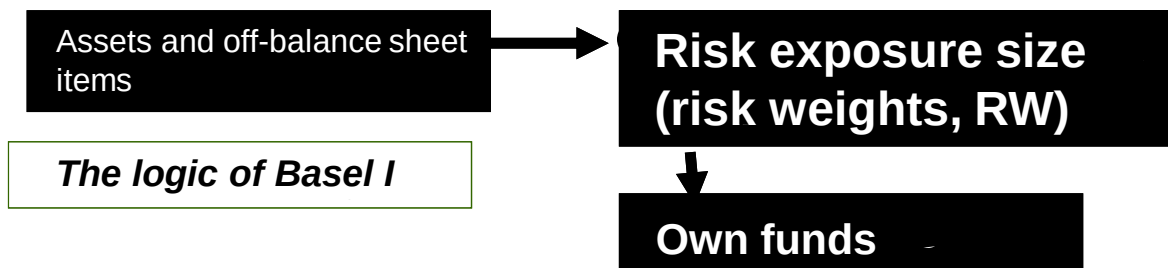
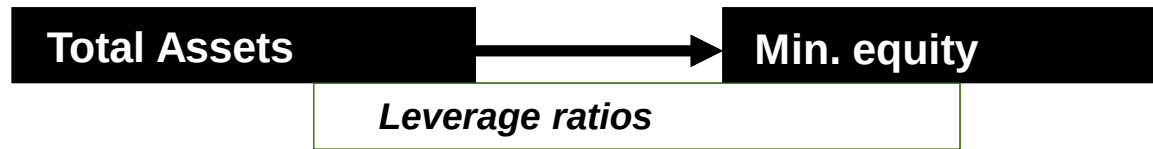
- The fundamental question for banks is therefore the measuring of **capital adequacy**: do they have enough capital to cover future losses?
- The possibility of loss exists regardless of the owner - can be best managed by the institution...
- ...but regulators can define the minimum capital required – based on maximum loss, ie VaR!

Regulators of the world, unite (in Basel)!

- Bank for International Settlements (BIS): bank for national banks, since 1930
- Basel Committee on Banking Supervision (BCBS):
 - Since 1974, organised by BIS, a forum of national banks and other supervisors
 - 13 BCBS countries: Belgium, Canada, France, Germany, Italy, Japan, Luxembourg, the Netherlands, Spain, Sweden, Switzerland, United Kingdom and United States
- One of the greatest achievements of the BCBS: the Basel Capital Accord of 1988 (at that time without serial number 1)

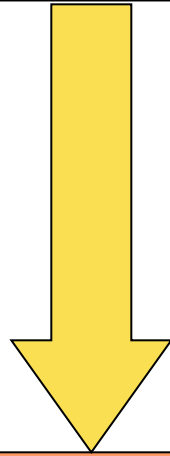


The logic of capital regulation



In the Basel I capital adequacy framework...

**Every 100 units of risk
should be backed by**



**Own funds of at least 8
units**



Own funds/Risk Weighted Exposure Amount (RWEA)

=Capital adequacy ratio > 8% (76.§ of the Credit Institutions Act)

What was "wrong" with the "Basel I" regulation ?

Not risk-sensitive

- 1) Ad hoc weights - not dependent on the bank's risk management and portfolio quality
- 2) „Rule based” regulation – „one-size-fits-all”, „Soviet razor”
- 3) The question "what is capital" is more important than "how much should capital be?,”
- 4) Deals only with credit risk

In short: regulatory capital had little to do with risk

With the emergence of the first VaR methodology (RiskMetrics) and VaR software (4.15), the concept of economic capital and quantitative risk management gained importance in the 1990s

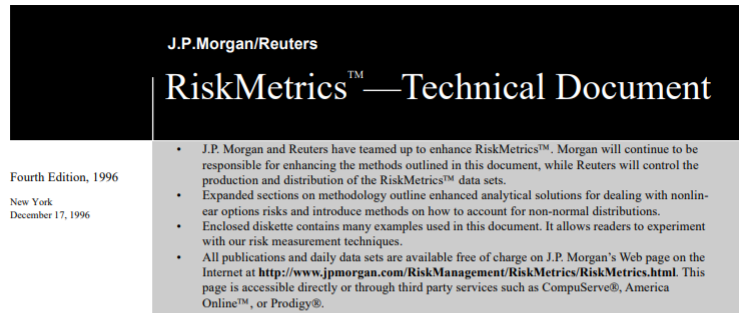


John Pierpont (J.P.)
Morgan (1837-1913)



Sir Dennis
Weatherstone

(J.P.Morgan CEO)
- 1994



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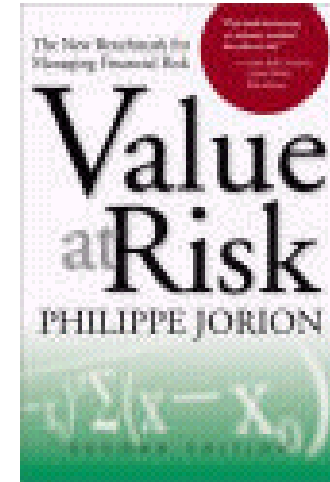
This *Technical Document* provides a detailed description of RiskMetrics™, a set of techniques and data to measure market risks in portfolios of fixed income instruments, equities, foreign exchange, commodities, and their derivatives issued in over 30 countries. This edition has been expanded significantly from the previous release issued in May 1995.

We make this methodology and the corresponding RiskMetrics™ data sets available for three reasons:

1. We are interested in promoting greater transparency of market risks. Transparency is the key to effective risk management.
2. Our aim has been to establish a benchmark for market risk measurement. The absence of a common point of reference for market risks makes it difficult to compare different approaches to and measures of market risks. Risks are comparable only when they are measured with the same yardstick.
3. We intend to provide our clients with sound advice, including advice on managing their market risks. We describe the RiskMetrics™ methodology as an aid to clients in understanding and evaluating that advice.

Both J.P. Morgan and Reuters are committed to further the development of RiskMetrics™ as a fully transparent set of risk measurement methods. We look forward to continued feedback on how to maintain the quality that has made RiskMetrics™ the benchmark for measuring market risk.

RiskMetrics™ is based on, but differs significantly from, the risk measurement methodology developed by J.P. Morgan for the measurement, management, and control of market risks in its trading, arbitrage, and own investment account activities. **We remind our readers that no amount of sophisticated analytics will replace experience and professional judgment in managing risks.** RiskMetrics™ is nothing more than a high-quality tool for the professional risk manager involved in the financial markets and is not a guarantee of specific results.



1996 Amendment (page 333-335)

- Implemented in 1998
- Requires banks to hold capital for market risk for all instruments in the trading book including those off balance sheet (This is in addition to the BIS Accord credit risk capital)

The Market Risk Capital

- The *[internal model approach]* capital requirement is

$$\max(\text{VaR}_{t-1}, m_c \times \text{VaR}_{\text{avg}}) + \text{SRC}$$

where m_c is a multiplicative factor chosen by regulators (at least 3), VaR is the 99% 10-day value at risk, and SRC is the specific risk charge for idiosyncratic risk related to specific companies. VaR_{t-1} is the most recently calculated VaR and VaR_{avg} is the average VaR over the last 60 days

Basel II

- Implemented in 2007
- Three pillars
 - New minimum capital requirements for credit and operational risk
 - Supervisory review: more thorough and uniform
 - Market discipline: more disclosure

Basel II Capital Requirements

- Risk weights based on either external credit rating (standardized approach) or a bank's own internal credit ratings (IRB approach)
- Recognition of credit risk mitigants
- Separate capital charge for operational risk

Basel II - a choice between simple and advanced measurement methods

- „Standard methods”:
 - Not based on an actual measurement of risk, but rather "approximate" regulatory capital weights are defined
 - Basel I's survival
- Advanced methods
 - Effective risk measurement and management
 - **This is the real Basel II, VaR-based methodologies**
 - **Must be authorised by supervisors**
- Credit risk
 - Standard method – regulatory RWs
 - **IRB, i.e. internal rating based method - based on internal models&Credit VaR formula**
- Market risk
 - Standard method: based on parameters given by the regulator
 - Internal model: application of own internal VAR model
- Operational risk:
 - BIA, i.e. basic indicator or STA, i.e. standard method
 - AMA i.e. Advanced Measurement Approach: actually measures risk



Credit risk measurement methods (pillar 1)

○ Standard method (STA)

- not subject to authorisation, simpler methodology (continuation of Basel I)
- guided by a number of legal passages
- less reflective of the specific risks of the institution

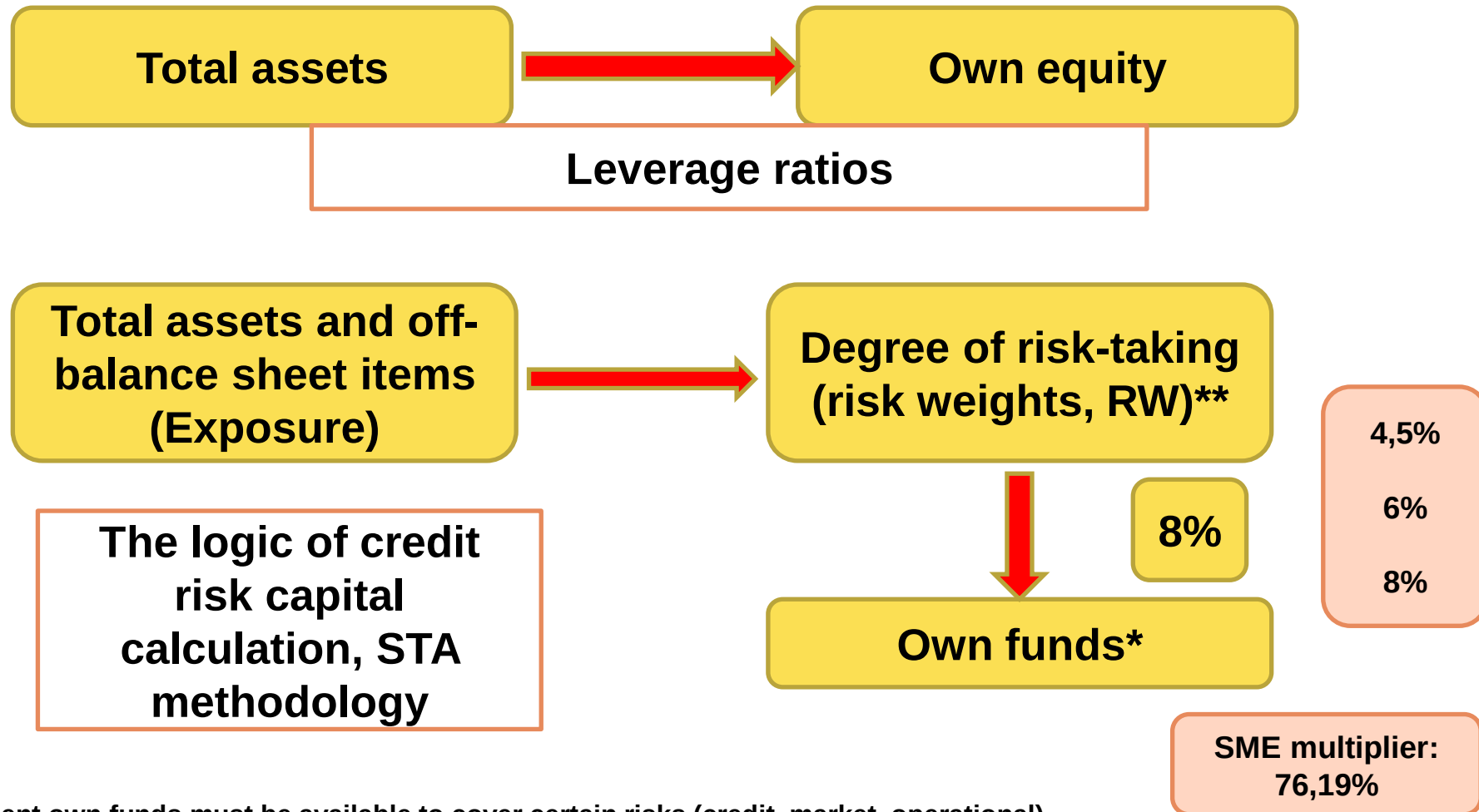
○ Internal **R**ating **B**ased (IRB) method

- Subject to supervisory authorisation
- guided by numerous legal passages AND numerous internal measurement methodologies
- VaR logic implemented through so-called capital functions
- results are better tailored to the institution through internal measurements

Principles of the standard method

- Essentially, it is based on the rating of debtors by an external rating agency and the ad hoc (arbitrary) Risk Weight (RW) assigned to it
- Key parameters:
 - External credit assessment institution (ECAI) rating (e.g. based on S&P, Moody's, Fitch rating classes 6 rating categories)
 - Risk weight assigned to rating category (RW)
- Techniques to mitigate risk (and capital requirements): Consideration of collateral (Credit Risk Mitigation, CRM)
- CRR Part Three, Title II, Chapter 2 (Standard method)

The logic of the standard methodology



*Sufficient own funds must be available to cover certain risks (credit, market, operational)

**For off-balance sheet items, the so-called conversion factor (CF) must also be taken into account

Segmentation by the standard method for determining risk weights

Regulatory exposure categories CRR/112.article

- central governments, central banks
- regional governments, local authorities
- public sector institutions
- multilateral development banks
- international organisations
- institutions (credit institutions and investment firms)
- companies
- retail claims
- exposures secured by a mortgage on real estate;
- non-performing exposures
- covered bonds
- items in the outstanding risk category
- exposures representing securitisation positions
- institutions, corporates with short-term credit ratings
- in the form of units or shares in collective investment schemes
- equity exposures
- other claims

- Within a segment, there may be further segmentations and different risk weights

-Fixed risk weights or external rating based classification (rating of Hungary or external rating of the counterparty behind the exposure)

-Special feature of standard segmentation: property is not collateral but an exposure class

Segmentation – classification examples

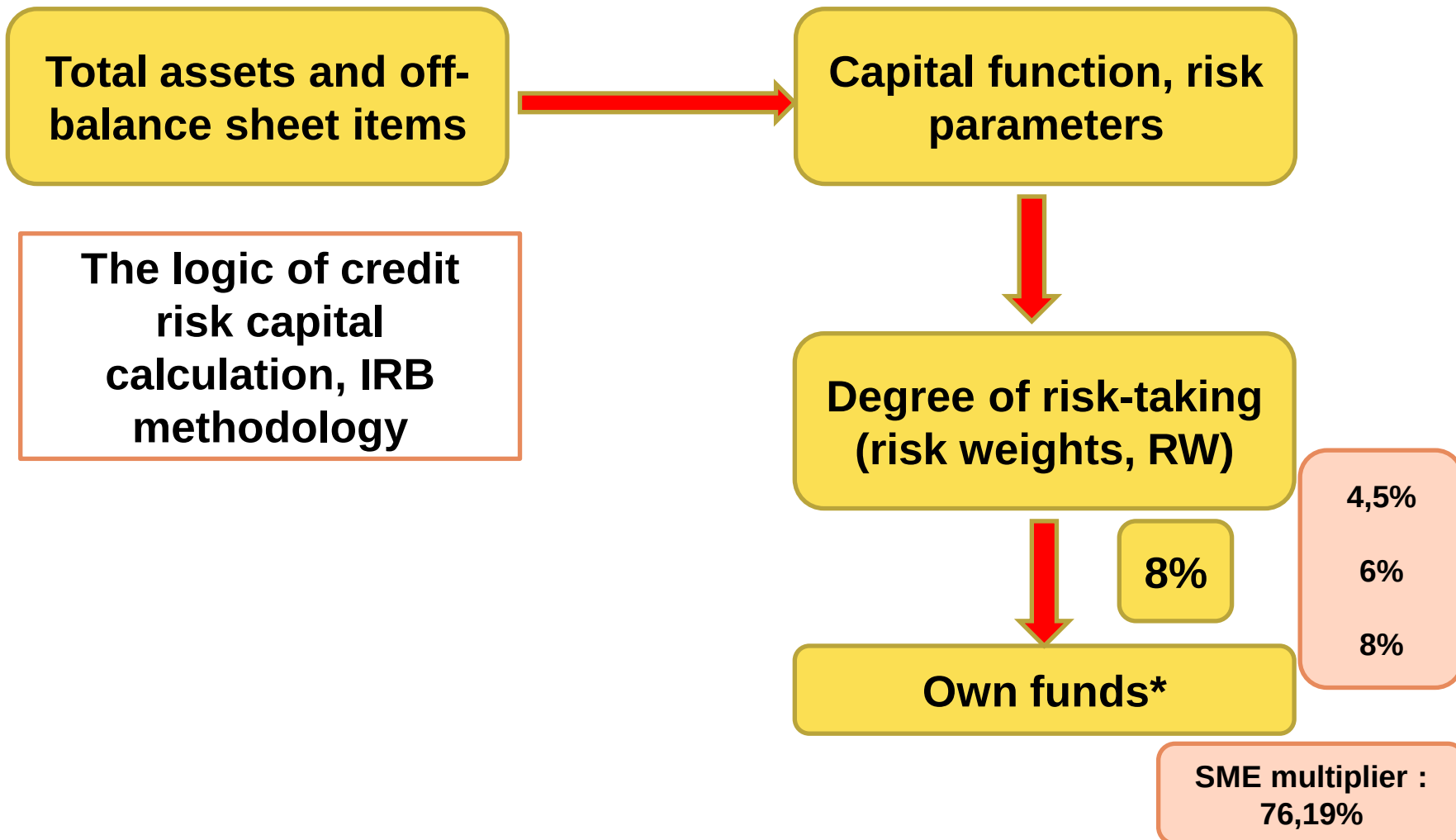
1.

Segment	Example - What might belong here?	Typical risk weight
<i>a) exposures to central governments or central banks, (CRR 114.article)</i>	government bond holdings	0% (with restrictions, transitional arrangement)
<i>b) exposure to regional governments or local authorities, (CRR 115. article)</i>	municipal bonds, claims on customers, exposures to the municipality	100%, in domestic currency 20%
<i>c) exposure to a public sector institution, (CRR 116. article)</i>	customer receivables from public or non-profit companies owned by the state or municipality (e.g. schools, universities, fire brigades, social institutions, public foundations, etc.)	State: 0%, municipal:100%
<i>d) exposure to a multilateral development bank, (CRR 117. article)</i>	claims on credit institutions, if there are any, e.g. EBRD,	0%
<i>e) exposure to an international organisation, (CRR 118. article)</i>	claims on counterparties, if there are any - 3 institutions: European Community, International Monetary Fund, Bank for International Settlements	0%

Segmentation – classification examples 2.

Segment	Example - What might belong here?	Typical risk weight
<i>f) Exposure to institutions, (CRR 119-121.articles)</i>	claims on credit institutions, exposures to equivalent financial corporations	20% (with restrictions), or 100%
<i>g) exposures to corporations, (CRR 122.article)</i>	receivables from customers on a residual basis (not included elsewhere)	100%
<i>h) exposure to the public, (CRR 123. article)</i>	exposures to counterparties may include exposures to natural persons or SMEs - up to a maximum of EUR 1 million exposure per counterparty group and granular portfolio	75%
<i>i) Exposure secured by a mortgage on real estate, (CRR 124. article)</i>	customer receivables secured by real estate are included in this part - taking into account the overcollateralisation requirement and other real estate compliance criteria (valuation, legal certainty and other strict requirements!)	residential property: 35% commercial property: 50%
<i>j) non-performing exposures, (CRR 127.article)</i>	Article 178 exposures on any claim (90+, unlikely to pay)	100-150%
<i>k) Exceptional high risk items (CRR 128. article)</i>	investments in venture capital companies; non-stock investments (private equity); speculative real estate finance	150%

The logic of the IRB methodology



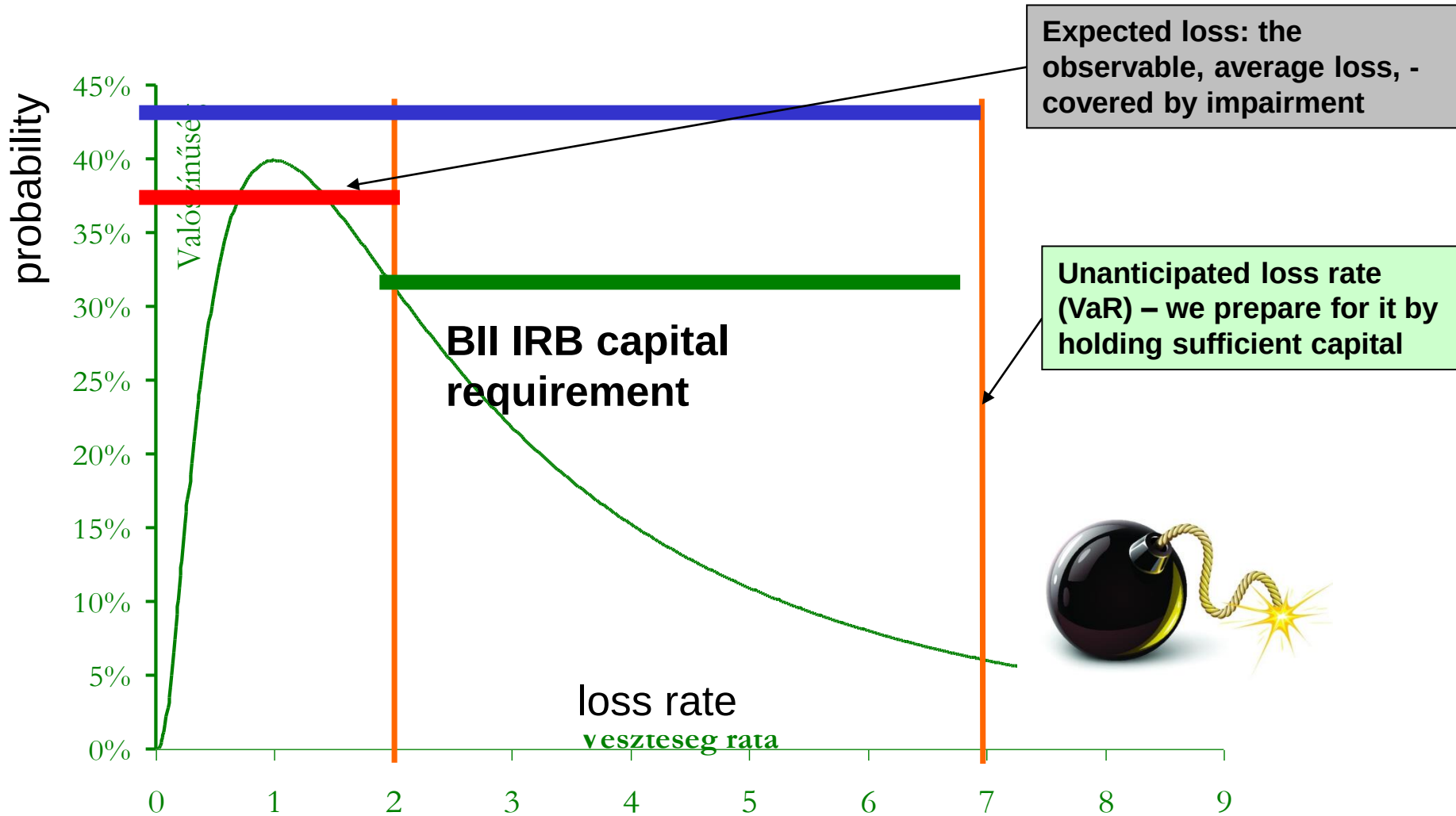
Characteristics of the IRB methodology

- **IRB** = Internal Rating Based
- Essentially a **statistical** approach **based on the rating** of debtors and transactions
- The amount of capital required is given by the **capital function**
- **Key parameters** that drive the capital function
 - **PD** (probability of **default**): debtor's default probability (statistical analysis of debtors) – **always own estimate**
 - **LGD** (loss given default): the magnitude of the loss at default (statistical analysis of workout)
 - **EAD** (exposure at default): total exposures (statistical analysis of off-balance sheet exposures – corresponding **CF** estimate)
 - **M** (maturity): average remaining maturity (duration type)
- Slightly different client/transaction **grouping** from the standard
- Risk mitigation options (**CRM**): expanding
- **There are hard limits to application, CRD**: Title V, Chapter 2, Section 3, Subsection 1, Article 84-89 + Annex VII; **Hkr. 24. §**-

LGD, CF, M values are given by Hkr.

<i>Own estimate</i>	FIRB	AIRB
Retail		PD, LGD, CF
States, institutions, companies	PD	PD, LGD, CF, M

The credit loss distribution & BII capital



VAR

The capital function

Expected loss (EL),
If $EL < \text{Impairment (I)}$, then
 $K = K + (EL - I)$ i.e. the
difference must also be
covered by capital!

- The legislation specifies the capital requirement function
- $K = (LGD \times N(x) - LGD \times PD) \times (1 + (M - 2.5) b) / (1 - 1.5 b)$
 - PD: probability of default
 - LGD: loss given default
- $X = [(1 - R)^{-0.5} \times G(PD) + (R / (1 - R))^{0.5} \times G(0.999)]$
- R: PD-dependent correlation with macro variable (given formula or value)
- M: duration type maturity variable (2.5 years normally)
- b: PD-dependent maturity adjustment parameter
- G(.): Inverse of normal distribution
- N(.): Normal distribution
- Risk-weighted assets (RWA) = $12.5 \times K \times 1.06 \times EAD$

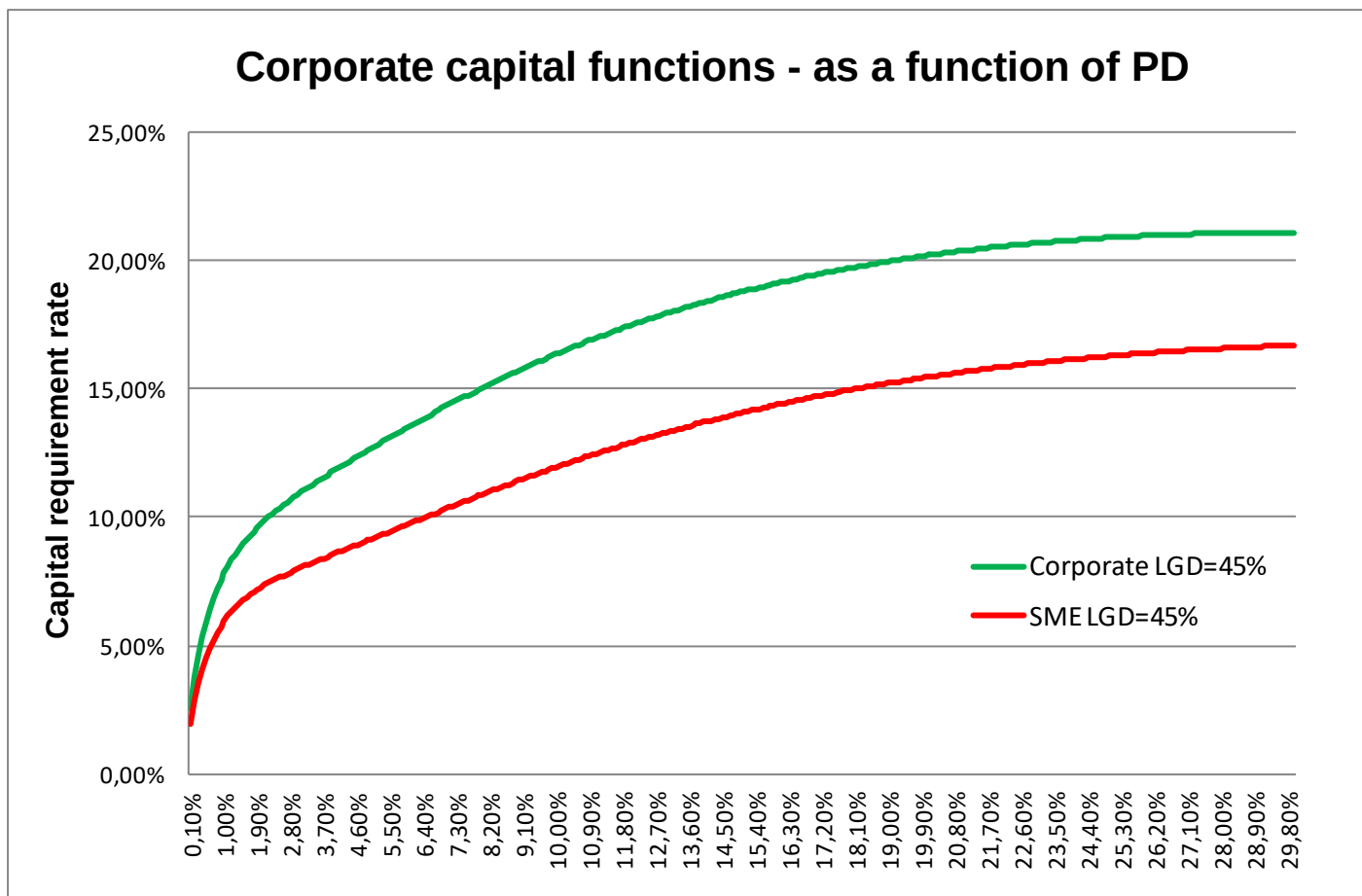
Maturity
adjustment
(base: 2.5 years)

"Scale factor"
raising the capital
level

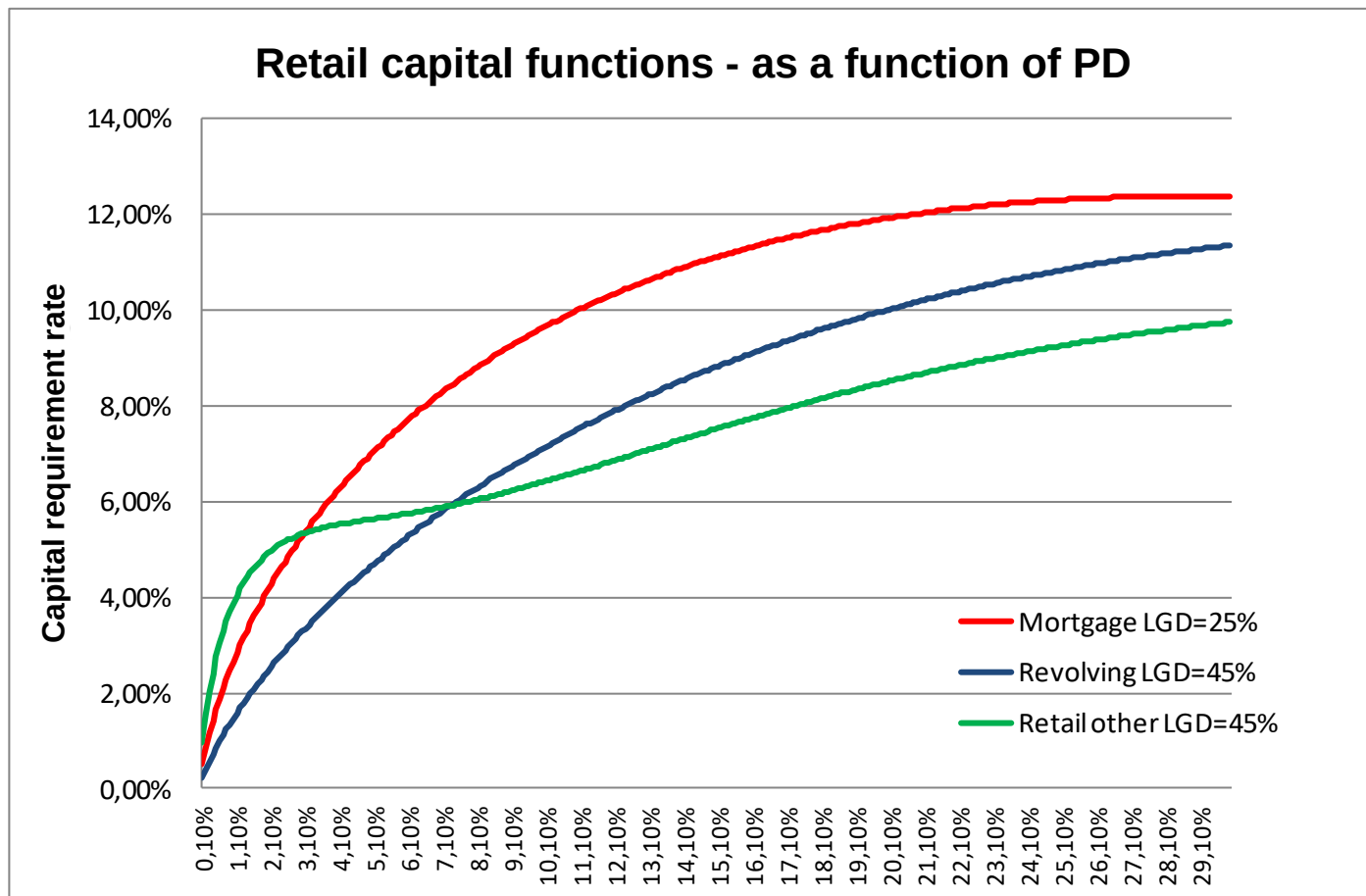
The 5+2 type capital function

- **Corporate** capital function: for countries, credit institutions, companies
 - Correlation factor varies between 12%-24%, the lower it is the higher the debtor's individual PD
 - There is a maturity correction
 - Very sensitive to PD growth under supervisory LGD
- Corporate capital function applied to **small and medium-sized enterprises**
 - Same as corporate, except that a correction factor has been added to the correlation, depending on turnover, with a maximum reduction of 4%
- Capital function of **retail mortgage claims**
 - Correlation coefficient (R) fixed: **15%**
 - Sensitive to the increase in LGD
- Capital function of **revolving retail credit**
 - Fixed, very low 4% correlation (sensitive to LGD)
- Capital function of **other retail claims**: also for small enterprises
 - 3 % - 16% correlation variation
- **Large financial sector firms and non-regulated financial firms**
 - The correlation coefficient is multiplied by 1.25

Example of corporate capital functions



Retail capital functions example



Basel III

- Capital Definition and Requirements
- Capital Conservation Buffer
- Countercyclical Buffer
- Leverage Ratio
- Liquidity Ratios
- Capital for CVA Risk

Types of Capital (Basel I)

- **Tier 1 Capital:** common equity, non-cumulative perpetual preferred shares
- **Tier 2 Capital:** cumulative preferred stock, certain types of 99-year debentures, subordinated debt with an original life of more than 5 years

Capital Definition and Requirements

- Three types:
 - Common equity Tier 1
 - **Additional Tier 1**
 - Tier 2
- Definitions tightened
- Limits
 - Common equity $> 4.5\%$ of RWA
 - Tier 1 $> 6\%$ of RWA
 - Tier 1 plus Tier 2 $> 8\%$ of RWA
- Capital levels implemented
- Phased implementation of capital definition stretching to January 1, 2018

Capital Conservation Buffer

- Extra 2.5% of common equity required in normal times to absorb losses in periods of stress
- If total common equity is less than 7% (=4.5%+2.5%) dividends are restricted
- Phased in between January 1, 2016 and January 1, 2019

Countercyclical Buffer

- Extra equity capital to allow for cyclicalities of bank earnings
- Left to the discretion of national regulators
- Can be as high as 2.5% of RWA
- Dividends restricted when capital is below required level
- Phased in between January 1, 2016 and January 1, 2019

Leverage Ratio

- Ratio of Tier 1 capital to total exposure (not risk weighted) must be greater than 3% (Higher in U.S. and UK)
- Exposure includes all items on balance sheet, derivatives exposures (calculated as in Basel I), securities financing exposures, and some off-balance sheet items
- To be introduced on January 1, 2018 after a transition period

Liquidity Risk Ratios

$$\text{Liquidity Coverage Ratio} = \frac{\text{High Quality Liquid Assets}}{\text{Net Cash Outflows for 30 day period}} \geq 100\%$$

for an acute 30 - day stress period (3 notch downgrade, partial loss of deposits, loss of unsecured wholesale funding, increased haircuts on secured funding, increased collateral requirements, drawdowns on lines of credit, etc)

$$\text{Net Stable Funding Ratio} = \frac{\text{Amount of Stable Funding}}{\text{Required Amount of Stable Funding}} \geq 100\%$$

for a period of longer term stress. Each category of funding (capital, deposits, etc) is multiplied by an available stable funding (ASF) factor to form numerator. Each category of required funding (assets, off - balance sheet exposures is multiplied by a required stable funding factor (RSF) to form denominator

The LCR and NSFR will be introduced on January 1, 2015 and January 1, 2018, respectively

USA vs European Implementation of Basel II

- In US Basel II applies only to large international banks
- Small regional banks required to implement “Basel 1A” (similar to Basel I), rather than Basel II
- European Union requires Basel II to be implemented by securities companies as well as all banks

Relevant eu regulation

The current Basel III regulatory standard was introduced in the European Union on 1 January 2014 and is implemented in the following two main pieces of legislation:

1) Regulation (EU) No 575/2013 (26 June 2013) on prudential requirements for credit institutions and investment firms, which is directly applicable in EU countries, **Capital Requirement Regulation** (hereafter **CRR**)

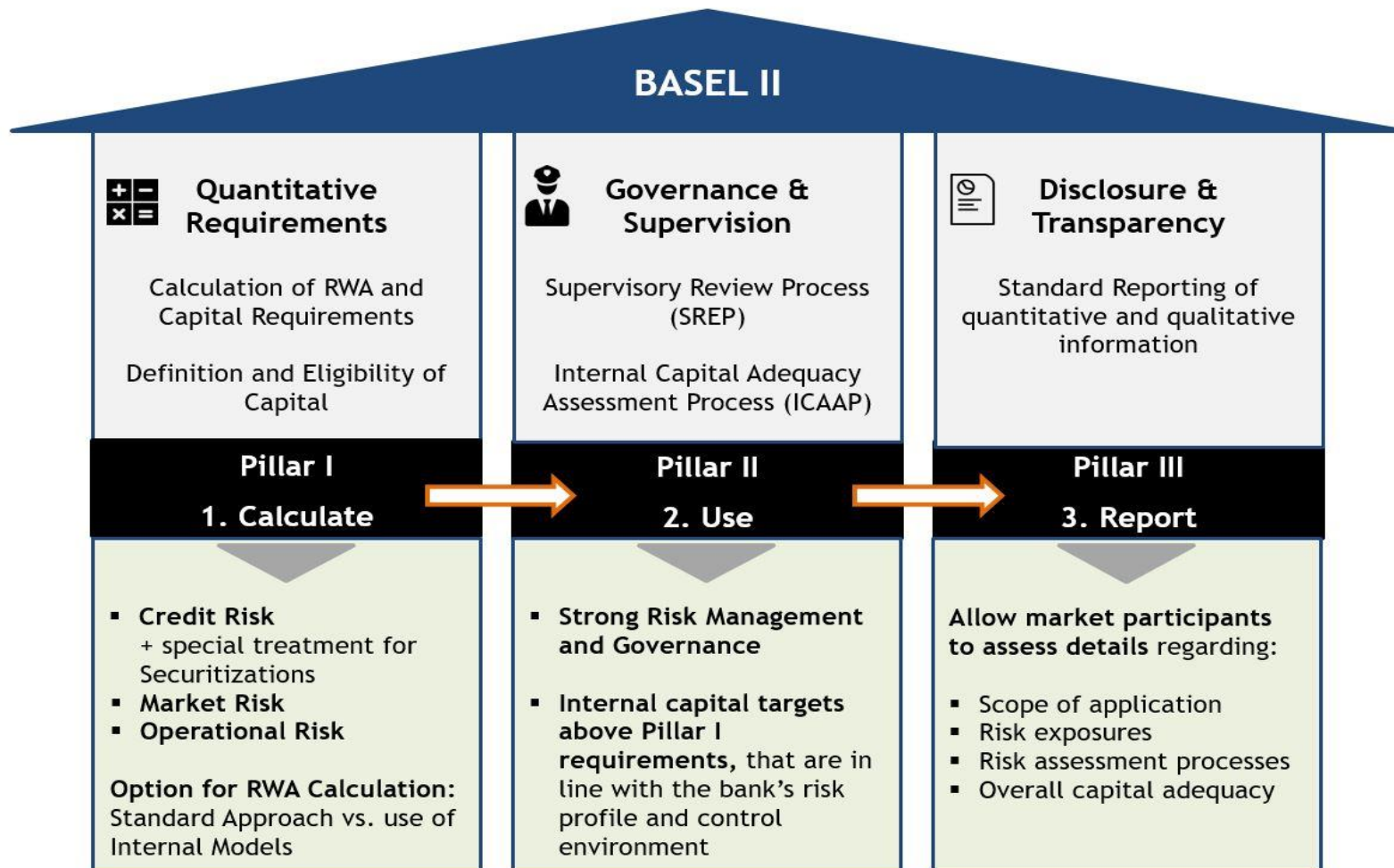
2) Capital Requirement Directive 2013/36/EU (26 June 2013) on access to the activity of credit institutions and the prudential supervision of credit institutions and investment firms, which must be implemented in national legislation **Capital Requirement Directive** (hereafter **CRD**)

CRR: Minimum capital
requirements, including credit
risk

CRD: conditions for the
authorization and operation
of banks

And Basel II (III) is not just the minimum capital requirements...

The 3 Pillars introduced under Basel II



bankárképző

TANÁCSADÁS ÉS OKTATÁS



**Thank you for your
attention!**

